

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

TITLE OF YOUR MASTER'S THESIS
MASTER'S THESIS

2026

TITLE FULL NAME, TITLE

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

TITLE OF YOUR MASTER'S THESIS
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Study Programme: Cognitive Science
Field of Study: Computer Science
Department: Department of Applied Informatics
Supervisor: title Full Name, title

Bratislava, 2026
title Full Name, title



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

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Typ záverečnej práce:

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Declaration

I hereby declare that the work presented in the master's thesis entitled "Title of Your Master's Thesis" is original and the result of my own research. Formulations and ideas taken from other sources are cited as such.

Bratislava, 2026

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Acknowledgments

Here you can thank your supervisor and/or other persons who somehow helped you with this thesis. You should acknowledge grants/projects in case your research was supported by any.

Abstract

Obligatory English abstract: 100–500 words; keep it formatted in one paragraph. Do not use abbreviations and other niche advanced terms here, which will be explained in the thesis later.

Keywords: first, second, third, (fourth, fifth)

Abstrakt

Obligatory Slovak abstract: should be the translation of the English one. Please ask a Slovak friend or an AI translator for help, if you are not native.

Kľúčové slová: jedno, druhé, tretie, (štvrté, piate)

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List of Abbreviations

LGN lateral geniculate nucleus

RGC retinal ganglion cell

Summary of Notation

Preamble to the list. Here you can put, for example, some general declarations or remove it completely. For instance: Capital letters (e.g., X) are used for random variables, sets, or matrices if boldfaced (e.g., $\mathbf{M}$). Lower-case letters denote scalars (e.g., x), and the values of random variables (e.g., $X = x$). Vectors are denoted by lower-case bold letters (e.g., $\mathbf{v}$).

General Notation

$\mathbb{R}$	real numbers
$\mathbb{N}$	non-negative integers
$\ \mathbf{u}\ _p$	L^p -norm of $\mathbf{u}$

Probabilistic Modeling

$\mathcal{U}[a, b]$	uniform distribution bounded by $[a, b]$
$\mathcal{N}(\mu, \sigma^2)$	normal distribution centered around μ with variance σ^2

Introduction

Chapter 1

Relatively Brief L^AT_EX Tutorial

In this chapter, we will go through the basic elements of L^AT_EX typesetting you may need for your thesis. If you are reading this chapter already rendered in the PDF, make sure to also follow its source code located in `sections/chapter.tex`.

Before we start, this template provides two basic commands to make your life a bit easier: `\todo` and `\tocite`. You can use `[TODO]` ... well, as a TO DO marker, and you can also use its parameterized version `\todo[something I have to do]` to leave comments for your future self. Similarly, you can `[reference needed]` to mark places where you would like to add references later.

1.1 Sectioning

One of the most automatic aspects of L^AT_EX is sectioning. At the top level, there are chapters, each automatically starting on a new page with a large header. You can start a new chapter anywhere by `\chapter{Chapter Name}`. Chapters can be split into sections (via `\section{Section Name}`) and further into:

1.1.1 Subsections

This text belongs to a numbered subsection.

1.1.1.1 Subsubsections

This text belongs to a numbered subsubsection.

Labeled Paragraphs This humble paragraph belongs to the subsubsection above. Note that the paragraph name is inline, rather than on its own line: *While questing once in noble wood of gray medieval pine, I came upon a tomb, rain-slicked, rubbed cool, ethereal; its inscription long vanished, yet still within its melancholy fissures –*

Note that all sections that are numbered automatically are also present in the table of contents. While this template allows numbering down to subsubsections, it is

generally not recommended to number them, as they tend to fragment the document too much. Paragraphs are also never numbered. If you would like to use headings without numbering, each of these commands has a starred version, which disables numbering and prevents the section from appearing in the table of contents:

Unnumbered Subsection

This text belongs to an unnumbered subsection.

Unnumbered Subsubsection

This text belongs to an unnumbered subsubsection.

1.2 Hyperlinking

In your thesis, you should link your sections, figures, tables, and equations together so readers can easily jump between different contexts. In L^AT_EX, this is done by assigning a label `\label{identifier}` to individual elements.

For instance, this section is labeled with `\label{sec:hyperlinks}`. It is best practice to use standardized prefixes for your identifiers to keep your source code organized:

- `ch:` for chapters (e.g., `ch:tutorial`)
- `sec:` for sections and subsections (e.g., `sec:hyperlinks`)
- `fig:` for figures (e.g., `fig:example`)
- `tab:` for tables (e.g., `tab:example`)
- `eq:` for equations (e.g., `eq:euler`)

Then, to refer to a labeled item, use `\cref{identifier}`. For example:

For more details, see Section 1.2.

`\cref` stands for *clever reference* and is a higher-level command that automatically determines the type of element (e.g., “Section”, “Figure”, or “Equation”) you are linking to.

Different elements, such as figures or tables, use different placements for the `\label{}` command; you will see exactly how in Sections 1.5 and 1.7.1 to 1.7.3.

1.3 Text Formatting

1.3.1 Fonts

In scientific writing, you can emphasize newly introduced terms, custom names, or important concepts. You can do this by italicizing the text via `\emph{text}` (for

emphasis) or `\textit{text}`:

this is important

Alternatively, you can use a bold font via `\textbf{text}`:

this is also important, but about as subtle as a sledgehammer

We generally avoid using underlining in academic writing, as it distorts the baseline of the font and can look quite ugly.

1.3.2 Dashes and Spaces

To maintain a bit prettier typography and avoid layout bugs, $\text{\LaTeX}$ provides specific syntax for different types of dashes and spacing:

- **Hyphen** (`-`): Used for compound words, e.g., `print-ready`.
- **En-dash** (`--`): Used for number ranges or as a parenthetical pause/digression, e.g., `pages 10--20`.
- **Em-dash** (`---`): Used as an unspaced parenthetical dash—like `this`—without spaces on either side.
- **Non-breaking space** (`~`): Use a tilde to prevent $\text{\LaTeX}$ from placing an awkward line break between two words that belong together (e.g., `Chapter~3`, `December~32`, or `Participant~P1`).

As you have probably noticed by now, you can make a new paragraph by leaving a separating blank line in your editor. Furthermore, $\text{\LaTeX}$ ignores any indentation or multiple spaces next to each other; it uses a space only as a signal that there is a new word following, but it prefers to manage the whitespace itself. Even if you start a new line but do not leave a blank line in your source code, $\text{\LaTeX}$ will continue typesetting in the same paragraph:

This is Paragraph 1.

This is Paragraph 2. Here, I will leave many spaces, but $\text{\LaTeX}$ does not care. Here, I even started on a new line in the code, but, still, $\text{\LaTeX}$ does not care and will just continue in Paragraph 2.

Only after I leave an empty line, it finally listens.

1.3.3 Abbreviations

This template is configured to allow you to define and use abbreviations out of the box. Usually, when you are introducing a longer term that will be used many times throughout the document, you also want to declare its abbreviation right next to it.

Optionally, you can also redeclare that abbreviation again on the first occurrence in every chapter, so the reader does not forget.

For example, perhaps you are introducing the lateral geniculate nucleus. First, define the abbreviation in `lists/acronyms.tex` via

```
\newacronym{identifier}{abbreviation}{full term},
```

like

```
\newacronym{lgn}{LGN}{lateral geniculate nucleus}.
```

Then, you can use it anywhere in the text using `\gls{identifier}`. You can also use `\glspl{identifier}`, if you want L^AT_EX to transform the abbreviation into the plural form:

The *lateral geniculate nucleus* (LGN) is a small, six-layered, ventral structure of the thalamus. LGN receives signal via the optic tract from the *retinal ganglion cells* (RGCs) and projects it via the optic radiation to the *primary visual cortex*.

1.3.4 Footnotes

You can provide side comments using footnotes, for example.¹ If you need to include a web link and do not have a formal citation for it, you can place it inside a footnote and wrap it in `\url{...}`.^{2,3}

1.4 Lists

If you need to format items in a list, you have two primary options. You can use the `itemize` environment for creating bullet-point lists:

The following *English* words are somehow special, and there are only about 80 such special words officially known in English:

- *bugsha*
- *cinq*
- *qinghaosu*
- *burqa*

¹A footnote should ideally be a full sentence or multiple proper sentences. You can use standard formatting elements *here* as well.

²<https://cogsci.fmph.uniba.sk/>

³You can also place multiple footnotes right next to each other; L^AT_EX will automatically separate them with commas.

- *yaqona*
- *qianzhousaurus*
- *shegel*

Or, you can use the `enumerate` environment for numbered lists:

These are the official names of the first six derivatives of position:

1. velocity
 - Also called *speed* in the scalar context
 - Usually denoted by $\vec{v}$ or $\boldsymbol{v}$
2. acceleration
3. jerk
4. snap/jounce
5. crackle
6. pop

1.5 Mathematics

Math in L^AT_EX can be typeset either inline or as displayed blocks. For inline math (formulas within a sentence), wrap your equations in single dollar signs, e.g., `$E = mc^2$`, which renders as $E = mc^2$.

For larger or more important formulas, use displayed equations. Ideally, your equations should be numbered and referable. For that, use the `equation` environment:

$$e^{i\pi} + 1 = 0. \tag{1.1}$$

You can then refer to it in your text using `\cref{eq:euler}`, which renders as Eq. 1.1. For a full reference and a list of all math symbols you can use, see *this guide*.

In case you use mathematical notation extensively, the template allows you to create a Summary of Notation. Navigate first to `lists/symbols.tex`. The formatting here is pre-configured, but the entries need to be created manually. In the code section **Groups**, you can define a notation group, as per the example there. Then, you can create a new symbol entry via

```
\glsxtrnewsymbol[description={...}, group={group identifier}]{sym:label}{notation}
```

These entries are purely static; they are not linked to the math in the document. You can also modify or remove the preamble of the summary.

1.6 Citations and Quoting

1.6.1 Citations

In L^AT_EX, citations are managed using a bibliography management system (such as BibT_EX or BibL^AT_EX). BibL^AT_EX is slightly more advanced, and since it allows us to use the full APA style for references, this template uses BibL^AT_EX.

BibL^AT_EX

In our case, all the references you use must be stored in `references.bib`. A reference entry can look like this:

```
@Article{Harnad1990,
  author      = {Harnad, Stevan},
  date        = {1990},
  journaltitle = {Physica D: Nonlinear Phenomena},
  title        = {The symbol grounding problem},
  doi          = {10.1016/0167-2789(90)90087-6},
  issn         = {0167-2789},
  number       = {1-3},
  pages        = {335--346},
  volume       = {42},
  publisher    = {Elsevier BV},
}
```

In the entry, `@Article` determines the type of the publication; in this case, it is a journal article. You can also have many others, e.g., `@InProceedings`, `@Book`, `@InBook`, `@PhdThesis`, `@Software`, etc. Afterward, in this case, `Harnad1990` is the identifier of the entry, so you can refer to it in the document. The rest of the fields provide details about the publication. You can provide many more fields than actually needed, because based on the bibliography style and the entry type, BibL^AT_EX will choose only the relevant ones and format them appropriately. The file can contain even unused references; only those that are used in the text will be rendered in the bibliography. Moreover, the references can be in any order in the file; BibL^AT_EX will sort the used ones according to the used bibliography style.

You should be able to download such a BibT_EX entry from any journal or conference proceedings website. If you cannot find any (often in the case of books; publishers' websites do not usually provide this option), you can always either create your own manually, or use one of many online tools or your favorite AI friend to format the BibT_EX entry for you. In case you are looking for a more complex solution, you can use, for example, the *JabRef* editor, which works directly with your `references.bib`.

Additionally, there are a few stylistic recommendations for your bibliography entries you might consider:

1. This template leaves the capitalization of the publication names mostly up to you. It is, though, advised to use title case for the names of articles, chapters, or any publications that are parts of larger works. For the names of journals, books, conferences and conference proceedings, theses, software, websites, or any other monographs, it is recommended to use sentence case. These names will also be rendered in italics, according to APA. Names should be correctly capitalized as well, of course, and a capital letter should follow a colon. Generally, to protect the capitalization in BibTEX entries, you can wrap the part you wish to be kept unchanged in `{...}`:

```
title = {{DiBS}: {D}ifferentiable {B}ayesian structure learning}
```

This template, however, preserves the capitalization of your entries without making automatic changes.

2. For the `date` field, leave just the year.
3. Use en-dashes for page ranges in the `pages` field.
4. In the `doi` field, use a pure DOI identifier, i.e., remove any prefixes such as `doi:`, `doi.org/`, or `https://doi.org/`. With the APA style, the DOI is the preferred identifier; no other will be displayed.
5. If you are using a DOI, remove any other field including web addresses, such as `url`.
6. You can include the journal or proceedings editors, but it is not necessary.
7. You can remove all unnecessary fields, e.g., `note`, `eprint`, `language`, `version`, `copyright`, `keywords`, etc. These will never be rendered.

Citing

Once we have a reference defined in `references.bib`, we can use it in citations. You can use *parenthetical citations* via `\citep{identifier}` when referencing some idea and the citation itself is not grammatically integrated in the sentence, for example:

The *symbol grounding problem* (Harnad, 1990) is indeed a problem.

Narrative citations via `\citet{identifier}`, on the other hand, are directly part of your sentence:

As Upper (1974) viscerally demonstrates, writing is sometimes hard; however, Thébaud et al. (2017) colorfully explore the enjoyable aspects of it.

Sometimes, we also want to specify the relevant page or chapter in the cited source. You can do it using the previous commands, for example, `\citep[p.~336]{Harnad1990}` or `\citet[Sec.~1.2]{Harnad1990}`, which will render as (Harnad, 1990, p. 336) and Harnad (1990, Sec. 1.2), respectively. This way, you can also refer to specific figures, tables, equations, etc. in the source. Note that the convention here is to abbreviate and capitalize; for example, *Section* as “Sec.” or *Figure* as “Fig.”, and also *page* as “p.”. You can also provide a page range; in such a case, you can write `pp.~42--50` (note the doubled `p` and the en-dash). Also notice that we are using here non-breaking spaces (`~`), as per Section 1.3.2.

1.6.2 Quoting

Typing standard keyboard double quotes (e.g., `"text"`) in L^AT_EX usually result in poorly formatted, backward-facing opening quotes (e.g., `"text"`). While native L^AT_EX allows formatting quotes using backticks and apostrophes (i.e., `'text'`), the cleanest way in this template is to wrap your text in `\enquote{...}` or `\textquote{...}`: “This is a quote.” The template also handles nested quotes: “She said, ‘Hello,’ and left.”

Usually, we also want to include a reference with a quote. According to the APA standard, you should also denote the specific page you took the quote from. Here, you can do it via `\textcquote[p.~123]{ref}{...}`: “According to connectionism, cognition is not symbol manipulation but dynamic patterns of activity in a multilayered network” (Harnad, 1990, p. 337).

For longer quotes (40+ words, according to APA), use a block quote via the `displayquote` environment, which automatically indents the text and formats it appropriately:

This is a block quote. It is useful for longer passages of text that you want to set apart from the main narrative flow of your thesis. You do not need to add quotation marks around it.

Similar to inline quoting, you can include references with block quotes via the `displaycquote` environment:

For the commercial aircraft, travel was related to trips the investigators were scheduled to take for business or personal reasons unrelated to the present study. Typically, passengers seated close to the study investigator (typically not known acquaintances) would be approached mid-flight, between the time of initial seating and time of exiting the aircraft. [...] All participants were asked whether they would be willing to be randomized to jump from the aircraft at its current altitude and velocity. [...] Participants were then instructed to jump from the aircraft after being provided their assigned device. (Yeh et al., 2018, pp. 1–2)

Note that we left out less important parts of the text using `\textelp{}` to make the quote more focused and shorter. Similarly, you can use `\textins{...}` to provide clarifying comments in quotes or to indicate a change in capitalization when you integrate a quote in your sentence:

As Toth et al. (2022, p. 16,263) note, “[f]rom a Bayesian perspective, an SCM can be viewed as a hierarchical data-generating process involving latent [i.e., exogenous] random variables” and locally specified causal mechanisms.

Finally, if you are quoting a text with an error by the original source, you should indicate it via `\textins{\emph{sic}}`: “In addition, Rui Long wants to thank, in particular, the patience, care and support from Panpan Mao over the passed [*sic*] years. Will you marry me?” (Long et al., 2018, p. 145).

1.7 Figures, Tables, and Algorithms

Elements like figures, tables, and algorithms are called “floats” because L^AT_EX dynamically calculates the best physical placement for them on the pages to avoid awkward blank spaces.

1.7.1 Figures

Figures are included using the `figure` environment. Always place your `\label{}` command after the `\caption{}` command to ensure that cross-referencing links correctly. The caption should be descriptive, explaining all visual elements and can span multiple sentences. If you mention a specific visual element, such as a dashed line, you can italicize it; in Fig. 1.1 we emphasized *boldface*. The caption will also be used automatically in the List of Figures as the name of the figure; however, captions are usually awfully long for this purpose, so you can define a short name which will be used instead via `\caption[short name]{long caption}`.

You can also adjust the size and the placement of the figure. Fig. 1.1 has its width set to 80% of the text width. While L^AT_EX likes to determine the placement on its own, you can express your preference via the parameters in `\begin{figure}[placement]`. You can use `h` (*here* – L^AT_EX will try to place the figure at the corresponding place or somewhere close), `t` (*top* – the figure will be glued to the top of the page), `b` (*bottom* – bottom of the page) and any combination of them, while the order determines the priority. If you get desperate, you can additionally use `!`, which will try to force L^AT_EX to comply; however, sometimes even that can be unsuccessful; in that case, try to move the figure to different places and sections.

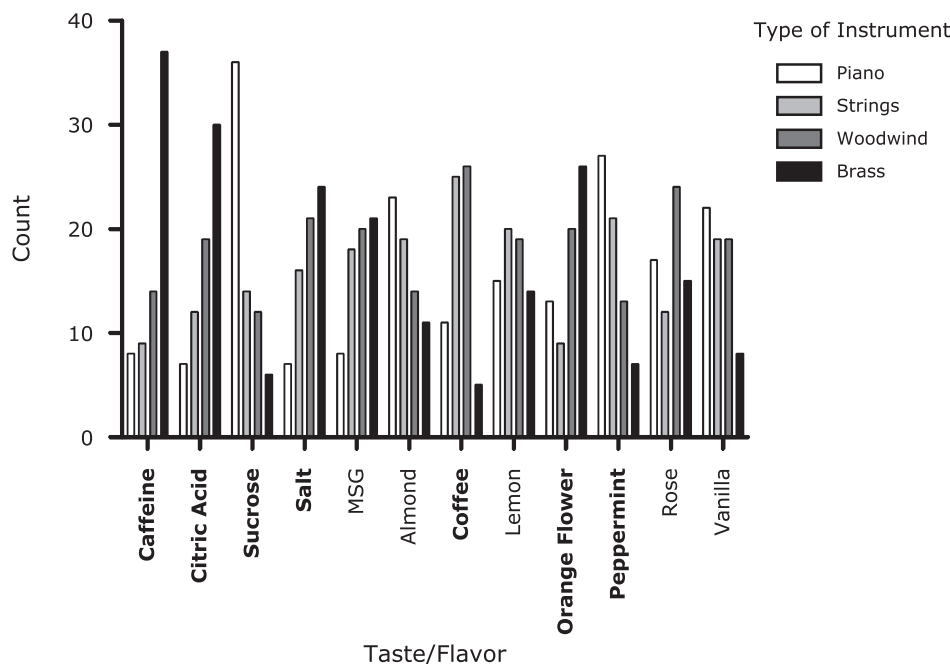


Figure 1.1: Musical instruments matched to the flavors. Stimuli for which there were significant instrument preferences are marked with *boldface* (Crisinel & Spence, 2010).

1.7.2 Tables

This template uses the *booktabs* table style, which avoids vertical lines and uses horizontal lines sparingly to make tables cleaner. Similar to the `figure` environment, you can define tables via the `table` environment with the same system of positional parameters.

Inside, the `tabular` environment defines the actual table: `\begin{tabular}{lll}` in the case of Table 1.1 declares a table with three columns aligned to the left (you can also use `c` for centering, and `r` for alignment to the right). In the table, you separate columns with `&` and lines with `\\`; any spacing around them is ignored. You usually do not need to input a table manually, as there are many web tools able to generate it for you from any kind of table format (e.g., *this one*).

Table 1.1: An example table. Notice that we place table captions above the table, whereas figure captions go below. Same as with figures, you can define a short name here as well.

Participant	Age	Condition
P01	22	Control
P02	25	Experimental
P03	21	Control

1.7.3 Algorithms

If your thesis involves pseudocode of some algorithms, you can typeset it using the `algorithm` environment provided, in our case, by the `algorithm2e` package,⁴ which has already been configured for this template. As with any pseudocode, you are free to use any combination of natural language, math notation, and programming elements to communicate the idea as efficiently and clearly as possible.

Same as with the previous floating environments, the `algorithm` environment works the same in terms of placement, captions, and labeling. If you use this environment anywhere in your document, the List of Algorithms will be generated; otherwise, it stays hidden.

Algorithm 1.1: Finding the maximum element in a sequence

Input: A finite sequence of numbers $A = \{a_1, a_2, \dots, a_n\}$

Output: The maximum element in the sequence

```

1  $max \leftarrow a_1$ 
2 for  $i \leftarrow 2$  to  $n$  do
3   if  $a_i > max$  then
4      $max \leftarrow a_i$ 
5 return  $max$ 
```

⁴For more details on the usage of this package, see *this tutorial*.

Conclusion

Bibliography

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Appendix A

Research Questionnaire

Appendix B

Implementation Source Code

Appendix C

Supplementary Figures